

Inhalational Burn

Section 1: Case Summary

Scenario Title:	Elderly Woman with Inhalational Burn
Keywords:	Inhalational burn, airway management, goals of care
Brief Description of Case:	A 70-year old female is brought to the ED after lighting herself on fire while trying to light a cigarette. She will have a 15% TBSA burn (upper anterior chest and neck only) with inhalational injury. Preparation for intubation should be an early priority. Her prognosis is poor but survivable and it will be key to discuss goals of care with the family before proceeding with intubation and further burn care. A difficult intubation should be anticipated but is not encountered in this case.

Goals and Objectives	
Educational Goal:	To review the management of an inhalational burn including an urgent goals of care discussion with a substitute decision maker
Objectives: (Medical and CRM)	<u>CRM Objectives:</u> • Task delegation (simultaneous patient management and family discussion) • Closed loop communication • Constructive intervention and mutual respect <u>Medical Objectives:</u> • Recognize signs and symptoms of an inhalational injury • Prognosticate severe burns using • Work with family to clarify goals of care quickly but compassionately • Quickly capture the airway prior to airway edema worsening
EPAs Assessed:	

Learners, Setting and Personnel	
Target Learners:	☒ Junior Learners ☐ Senior Learners ☐ Staff
	☐ Physicians ☐ Nurses ☐ RTs ☐ Inter-professional
	☐ Other Learners:
Location:	☒ Sim Lab ☒ In Situ ☐ Other:
Recommended Number of Facilitators:	Instructors: 1
	Confederates: 1-2 (EMS, RN, family member)
	Sim Techs: 1

Scenario Development	
Date of Development:	January 2, 2018
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Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: Helen Jones		Age: 70		Gender: female	Weight: 50 kg
Presenting complaint: Thermal Burn					
Temp: 37.1°C	HR: 110/min	BP: 145/87	RR: 22/min	O ₂ Sat: 98%	FiO ₂ : 100%
Cap glucose: 5.4 mmol/L			GCS: 13 (E3 V4 M6)		
Triage note: EMS called after accidentally lighting herself on fire after trying to light a cigarette. Was on fire for several minutes before staff arrived and were able to extinguish the fire. Patient conscious on EMS arrival.					
Allergies: None					
Past Medical History: Parkinson's Hypothyroidism Hypertension			Current Medications: Ramipril Carbidopa-Levodopa		
Social History: Lives in Assisted Living due to poor mobility from Parkinson's					

Section 2B: Extra Patient Information

A. Further History	
Single victim, burns apparent but no other injuries noted Has been conscious with EMS throughout. Goals of care unclear at present. No further history obtained.	
B. Physical Exam	
General Status: Eyes closed unless spoken to, states she is in pain but otherwise provides nonsense answers to questions. Thin, frail woman.	
Cardio: Heart sounds normal, no EHS	Neuro: Decreased level of consciousness, GCS 13
Resp: Wheezy on initial assessment, no stridor	Head & Neck: soot in mouth, burns on neck
Abdo: Soft, non-tender	MSK/skin: 2 nd degree burns to neck and parts of chest
Other: N/A	



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Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin - adult, female
☐ Standardized Patient
☐ Task Trainer
☐ Hybrid
B. Special Equipment Required
Standard airway equipment Difficult airway equipment
C. Required Medications
RSI meds Awake intubation meds Analgesia
D. Moulage
Adult sized female mannequin. If possible, moulage make up to simulate burns on upper chest and neck, soot around mouth.
E. Monitors at Case Onset
☒ Patient on monitor with vitals displayed ☐ Patient not yet on monitor
F. Patient Reactions and Exam
<i>Include any relevant physical exam findings that require mannequin programming or cues from patient (e.g. – abnormal breath sounds, moaning when RUQ palpated, etc.) May be helpful to frame in ABCDE format.</i> Wheezing until neb is finished Hoarse voice

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Section 4: Confederates and Standardized Patients

Confederate and Standardized Patient Roles and Scripts	
Role	Description of role, expected behavior, and key moments to intervene/prompt learners. Include any script required (including conveying patient information if patient is unable)
Paramedic	Gives initial description of patient to team: “This is Helen, she is a 70 y o female is brought to the ED after lighting herself on fire while trying to light a cigarette. She has a history of Parkinson’s so she had difficulty putting out the fire on herself, so she was on fire for several minutes before staff were able to put it out. She was wheezy on scene which has improved with a nebulizer en route. Her vitals have been: HR 110, BP 145/87, Temp 37.1 °C, RR 22, O2 93% RA (98% with neb). Her daughter/son (depending on gender of confederate) is on her way. I believe she is bringing her advance care directive.”
Resus Nurse	Hooks patient up to monitors, provides additional vitals if requested (Blood sugar 5.7, Temperature 37.1°C). When patient is examined, nurse will describe the patient as having 2 nd degree burns to parts of her chest and neck and when the patient’s mouth is inspected, they will describe intra-oral soot and singed eyebrows.
Daughter/Son	Will enter the room after initial trauma assessment with the advanced care directive stating “Full Code”. They will be distraught at the situation, state that the directive was signed a couple years ago when their mother was more ‘with it’ and living at home. She’s become much more frail lately. Asks what the likelihood of recovery will be. Eventually commits to whatever the document says and that they will contact the rest of the family.

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Section 5: Scenario Progression

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Baseline State Rhythm: sinus tach HR: 110/min BP: 145/87 RR: 20/min O ₂ SAT: 98% T: 37.1°C Gluc: 5.4 mmol/L GCS: 13	Confused, diffuse wheeze	<u>Expected Learner Actions</u> ☐ IV access/labs, monitors ☐ Determine weight ☐ Recognition of possible inhalational injury ☐ Recognizes need for early intubation and prepares for such ☐ Recognizes need to identify goals of care ☐ Log roll patient ☐ Continue Ventolin neb	<u>Modifiers</u> - Wheeze will resolve once neb is finished <u>Triggers</u> - Paramedic states the son/daughter is coming in with her advance care directive	
2. Goals of care discussion Wheeze resolved Son/daughter arrives with advanced care directive	Same level of confusion	☐ Assigns a team member to manage patient OR to talk to son/daughter about advance care directive and goals of care ☐ Burn Centre consult ☐ Determines rough prognosis of this burn/inhalational injury to be survivable but poor	<u>Modifiers</u> - N/A <u>Triggers</u> - After discussion with family, they want to proceed with therapy/intubation	
3. Intubation		☐ Prepares drugs for intubation ☐ Verbalizes Plan A, B, C for capturing airway ☐ Consider anesthesia/ICU for airway assistance ☐ Post-intubation care ☐ Consults ICU	<u>Modifiers</u> - Hemodynamics and oxygenation good post-intubation - Easy intubation <u>Triggers</u> - Call ICU → End of Case	<u>Airway plan should consider:</u> - potential narrowed airway - airway edema - need for smaller ETTs



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Appendix A: Laboratory Results

No labs given during case

Appendix B: ECGs, X-rays, Ultrasounds and Pictures

No imaging given during case

Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Review of Objectives:

Education Goal: To review the management of inhalational injury including an urgent goals of care discussion with a substitute decision maker

CRM Objectives:

- Task delegation (simultaneous patient management and family discussion)
- Closed loop communication
- Constructive intervention and mutual respect

Medical Objectives:

- Recognize signs and symptoms of an inhalational injury
- Prognosticate severe burns using
- Work with family to clarify goals of care quickly but compassionately
- Quickly capture the airway prior to airway edema worsening

Sample Prompting Questions:

- 1) Did you feel that your team leader had control over the room? Did he or she ask for input?
- 2) How do you navigate goals of care and advance care directives when there is a time sensitive illness/injury?
- 3) How do you determine prognosis in a burn?
- 4) Is succinylcholine safe to use in a burn patient? Why or why not?

Common Questions/Concerns:

- 1) Burn prognosis is not a perfect science. The modified Baux score can serve as a guide. Age + TBSA % + presence/absence of inhalational injury (+ 17). Historically the number correlated to mortality. So in this case, 70 years old + 20% TBSA + 17 = 107% mortality!! The reality is that burn care has improved tremendously and modified Baux scores under 140 are all potentially survivable. Her predicted mortality is still very high, but it is not futile.
- 2) Succinylcholine is safe in an acute burn. The burn process is thought to cause a loss of neuromuscular junction receptors, leading to up-regulation of those receptors. Once up-regulated, the depolarization can lead to transient hyperkalemia and subsequent cardiac arrhythmias. The process of up-regulation takes 24-48 hours, so in the acute phase, succinylcholine is safe to use for intubation.

References

1. Sheridan RL. 2016. Fire-Related Inhalation Injury. *N Engl J Med*. 375(5):464-9
2. Billington, M. 2016. Modern Day Burn Resuscitation. *EMdocs.net*. Retrieved from <http://www.emdocs.net/modern-day-burn-resuscitation-moving-beyond-parkland-formula/>

